Coffee Shop Sales Analysis (Jan to Jun 2023)

An interactive Excel dashboard on revenue, footfall, peak hours and product mix across three New York coffee stores

Project type	Portfolio project built on a publicly available practice dataset (not client work)
Tools	Microsoft Excel: Power Query, Data Model, Pivot Tables, Pivot Charts, Slicers, dashboard design
Data	149,116 transactions | 3 stores | 9 product categories | 1 Jan to 30 Jun 2023 | values in USD
Author	Abhijit Sharma, B.Tech Computer Science (Data Science)

1. Executive summary

This project analyses six months of point-of-sale data from a coffee-shop business with three New York stores: Astoria, Hell's Kitchen and Lower Manhattan. Sales totalled **$698,812** across 149,116 transactions, with an average bill of **$4.69** and 1.44 items per transaction. Monthly revenue **doubled from January to June (+104%)**, **46% of revenue arrives between 7 and 10 AM**, coffee and tea make up two-thirds of sales, and the three stores perform almost identically. This report explains the problem, the build process, the findings and the actions they support.

$698,812	149,116	$4.69	1.44
Total sales	Footfall (transactions)	Average bill per transaction	Items per transaction

2. Problem statement

The owner of a small coffee chain wants to know **when customers buy, what they buy, and how the three stores compare**, so staffing, stock and promotions can be planned with evidence instead of guesswork. The raw data is 149,116 individual till transactions, which is far too granular to read. The goal of this project is a single dashboard that answers the main operational questions and can be filtered by day of week and month.

Business questions the dashboard answers

1. When are the peak hours, and how does footfall vary by day of the week?
2. How is revenue trending month by month?
3. How do the three store locations compare on footfall and sales?
4. What is the average bill, and how many items does a customer buy per visit?
5. Which product categories and product types earn the most?
6. How are orders split by cup size (small, regular, large)?

3. Dataset

Source	Maven Analytics public "Coffee Shop Sales" sample dataset (practice data, not real client data)
Records	149,116 transactions, each with date, time, quantity, unit price, store, product category, product type and product detail
Coverage	3 stores, 9 categories, 29 product types, 80 products; 1 January to 30 June 2023
Quality checks	No missing values; every transaction ID is unique; quantity × unit price equals the sales total exactly

4. Solution process

1. **Import and shape the data with Power Query.** Loaded the transactions into a Power Query query named Transactions and into the Excel Data Model, so the pivot tables work on a clean, refreshable source.
2. **Add derived fields.** Created Total Bill (quantity × unit price), Size (separated from the product detail text), Month Name, Day Name, Hour, Month number and Day of Week to support time-based analysis.
3. **Summarise with Pivot Tables.** Built ten pivot tables: items sold by hour, revenue by month, orders by size, footfall and sales by store, revenue by category, top product types, footfall by weekday, and the KPI measures.
4. **Choose the right chart for each question.** A line chart for the hourly pattern, a pie chart for size mix, and column charts for stores, weekdays, categories and product types.
5. **Create KPI cards.** Total sales, footfall (count of transactions), average bill (sales ÷ transactions) and items per transaction.
6. **Add slicers and assemble the dashboard.** Connected Day and Month slicers to every chart and arranged everything on one sheet in a consistent coffee-themed colour palette.

5. The dashboard

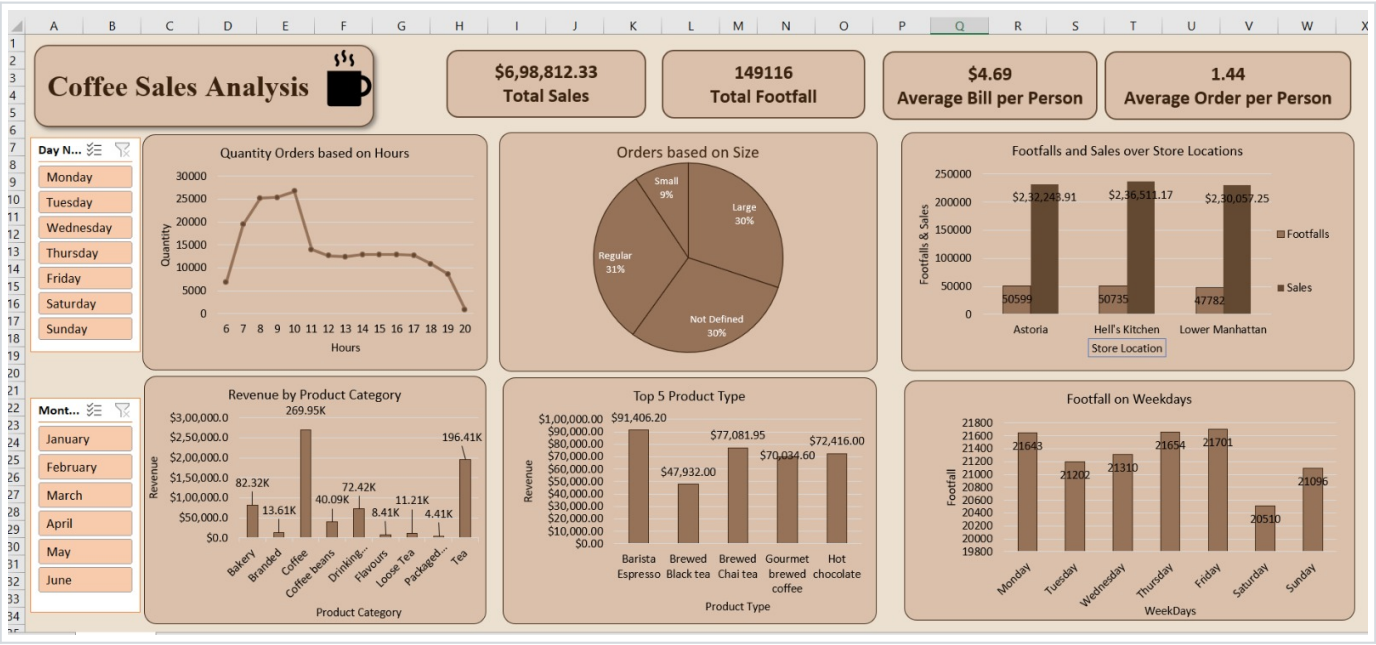


Figure 1. Coffee Sales Analysis dashboard: four KPI cards, six charts and Day and Month slicers.

How to use it: select one or more days or months in the left slicers; the KPI cards and all six charts update together. Use the funnel icon to clear a filter.

6. Key findings

1. **Revenue doubled in six months, driven by volume rather than price.**
Monthly revenue grew from **$81,678 in January to $166,486 in June (+103.8%)**, while transactions rose from 17,314 to 35,352. The average bill stayed flat at $4.65 to $4.72 every month, so growth came from more customers, not higher prices. February's dip (-6.8%) is a calendar effect: it has 28 days, and its daily average ($2,719) is actually above January's ($2,635).

2. **A strong morning rush defines the day.**
The hours from 8 to 10 AM sell **36.0% of all items**, and 7 to 10 AM brings **45.8% of revenue** in just 4 of the 15 trading hours. Items sold peak at 10 AM (26,713) and drop by about 47% at 11 AM (14,035). From 11 AM to 5 PM, demand is steady at roughly 12,500 to 14,000 items per hour, then tapers after 6 PM.

3. **The three stores perform almost identically.**
Hell's Kitchen ($236,511, 33.8%), Astoria ($232,244, 33.2%) and Lower Manhattan ($230,057, 32.9%) are within 3% of each other. Lower Manhattan reaches the same revenue with fewer transactions because each customer spends more: **$4.81 per bill and 1.50 items**, versus $4.59 to $4.66 and 1.40 to 1.41 at the other two stores.

4. **Coffee and tea are two-thirds of the business.**
Coffee ($269,952) and Tea ($196,406) contribute **66.7% of revenue**; Bakery adds 11.8% and Drinking Chocolate 10.4%. The top five product types (Barista Espresso, Brewed Chai tea, Hot chocolate, Gourmet brewed coffee, Brewed Black tea) earn 51.4% of sales; Barista Espresso alone earns $91,406 (13.1%).

5. **Day of week barely matters; time of day does.**
Daily footfall ranges only from 20,510 (Saturday) to 21,701 (Friday), a spread of about 5.6%. Saturday is the weakest day at 5.5% below Friday. Planning should follow the hour of the day rather than the weekday.

6. **Regular and Large sizes dominate; Small is a niche.**
Regular (30.7% of transactions) and Large (30.1%) dominate, while Small is only 9.3%. A further 29.9% of transactions are "Not Defined" because bakery, coffee beans, merchandise and flavour items have no size. Small is offered only on coffee drinks.

7. **Supplementary: some products add volume but little revenue, others are high-value baskets.**
Computed from the same dataset: Flavours are 4.9% of items sold but only 1.2% of revenue, while a coffee-beans purchase averages **$22.87** compared with $4.69 for the average bill.

7. Supporting figures

Monthly revenue

Month	Revenue	Transactions	MoM
January	$81,678	17,314	-
February	$76,145	16,359	-6.8%
March	$98,835	21,229	+29.8%
April	$118,941	25,335	+20.3%
May	$156,728	33,527	+31.8%
June	$166,486	35,352	+6.2%

Items sold by hour of day

Hour	Items sold
6 AM	6,865
7 AM	19,449
8 AM	25,197
9 AM	25,370
10 AM	26,713
11 AM	14,035
12 to 5 PM (per hour)	12,439 to 12,923
6 PM	10,826
7 PM	8,595
8 PM	880

Revenue by category

Category	Revenue	Share
Coffee	$269,952	38.6%
Tea	$196,406	28.1%
Bakery	$82,316	11.8%
Drinking Chocolate	$72,416	10.4%
Coffee beans	$40,085	5.7%
Branded	$13,607	1.9%
Loose Tea	$11,214	1.6%
Flavours	$8,409	1.2%
Packaged Chocolate	$4,408	0.6%

Store comparison

Store	Revenue	Txns	Avg bill	Items/txn
Astoria	$232,244	50,599	$4.59	1.40
Hell's Kitchen	$236,511	50,735	$4.66	1.41
Lower Manhattan	$230,057	47,782	$4.81	1.50

8. Recommendations

1. **Staff and stock for the 7 to 10 AM rush.** Nearly half of revenue arrives in these three hours; prepare espresso and bakery stock beforehand and consider mobile pre-ordering to shorten queues.
2. **Create an afternoon offer.** From 11 AM to 5 PM demand is steady but about half the morning peak; a tea and bakery combo could lift this slower window.
3. **Run a weekend promotion.** Saturday is the weakest day; an offer there can narrow the gap with Friday.
4. **Raise items per transaction.** At 1.44 items per visit there is room to bundle a drink with a bakery item (average bakery bill $3.61). Study why Lower Manhattan reaches 1.50 and copy the practice at the other two stores.
5. **Promote high-value baskets.** Coffee beans average $22.87 per purchase; a checkout prompt or sampling could grow this small but valuable line.
6. **Extend the analysis.** Collect a full 12 months to separate seasonal growth from genuine expansion, and add cost data to measure profit instead of revenue.

9. Data notes and limitations

- **Footfall is a proxy.** It equals the number of transactions; the data has no customer ID, so one person buying twice counts twice.
- **Six months only.** The January to June growth could be seasonal or due to expansion; the data cannot tell the two apart.
- **Revenue, not profit.** No cost or margin data is available.
- **"Not Defined" size** means the item has no size (for example scones, beans, merchandise), not missing data.
- **Closing time is inferred.** Only 880 items sell in the 8 PM hour, which suggests stores close around that time.

10. Skills demonstrated

> Power Query data import and shaping | Excel Data Model | Pivot Tables and Pivot Charts | Slicers | KPI card design | Time-series and peak-hour analysis | Dashboard layout and storytelling | Business recommendations

Note: this is an independent portfolio project using a public practice dataset. All figures in this report were verified against the dataset in the accompanying Excel workbook.